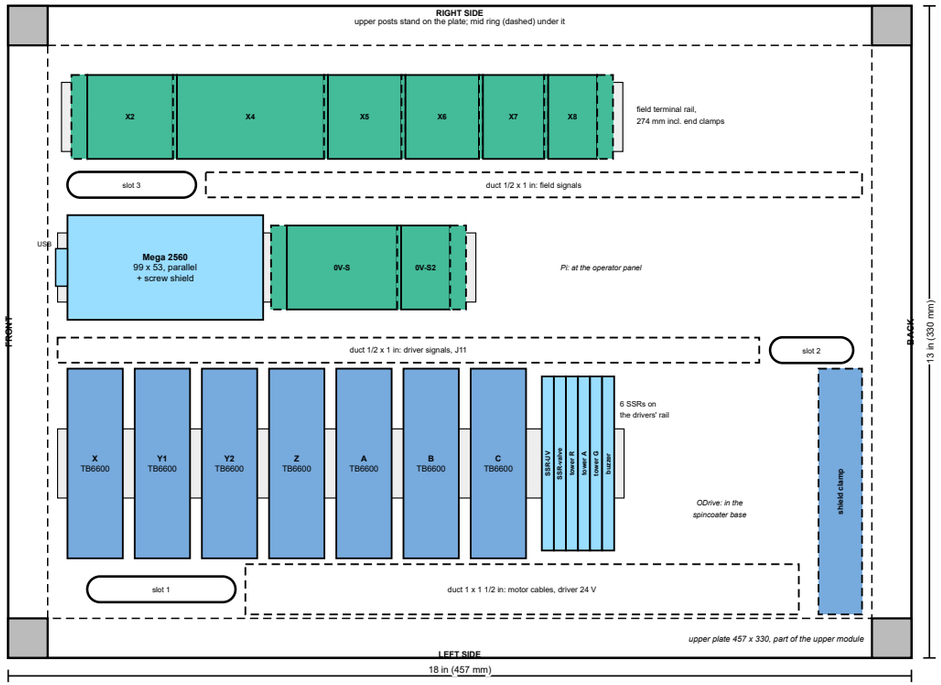
RMR control panel: packing study, 18 x 13 x 11 in

Two decks in a 2020 frame, both on 1/16 in aluminum plates: lower deck for 110 VAC and 24 V power, upper deck for drives and logic; acrylic walls. Scale 1:3.5 when printed at 100 %.

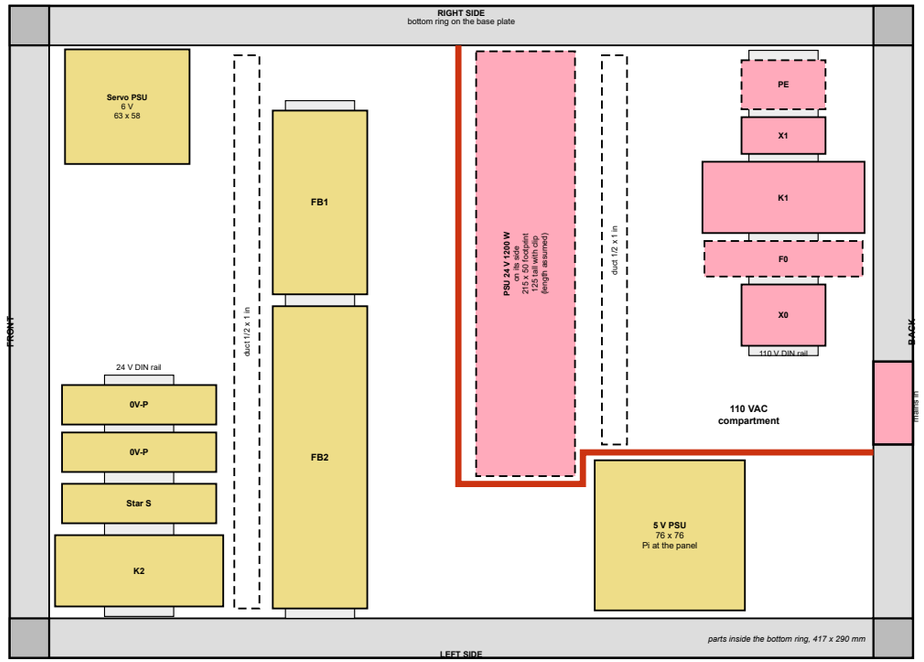
A Upper deck, seen from above

1/16 in aluminum plate on the mid ring; parts at true size



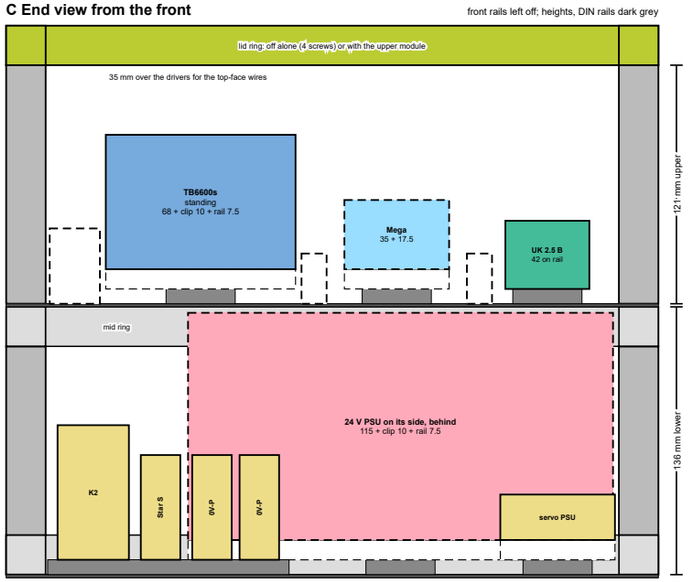
B Lower deck, upper deck lifted off

1/16 in aluminum base plate, bonded to PE



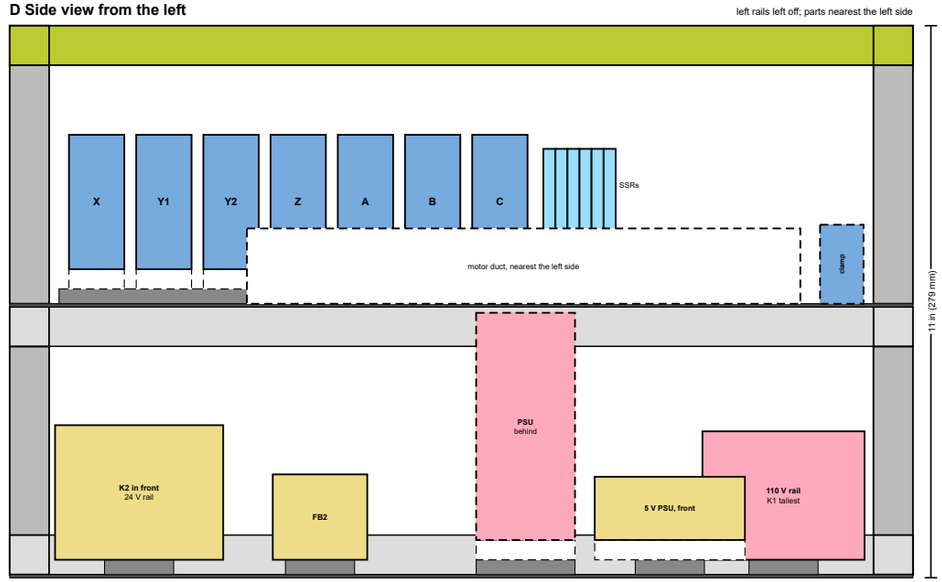
C End view from the front

front rails left off; heights, DIN rails dark grey



D Side view from the left

left rails left off; parts nearest the left side



Zones

- 110 VAC and the PSU
- 24 V distribution and DC-DC
- drives
- logic and SSRs
- field terminals
- 2020 frame (posts darker)
- lid ring, lifts off
- 1/16 in aluminum plate, edge-on
- DIN rail, 7.5 tall (elevations)
- solid outline: measured or datasheet
- dashed: size assumed, measure

Mounting

Every part stands on a 7.5 mm DIN rail; parts without DIN feet add a printed clip, 10 mm. TB6600s stand on their 96 x 28 base, ears 88 apart across the rail, fins vertical. The 24 V PSU stands on the face with the four holes (200 x 32.5), rail across its length, as in PSU DIN mount.SLDPRT. The Mega lies parallel to its rail, so the screw shield's terminals face both ducts.

Height budget (mm)

base plate 1.6, rail 7.5 + clip 10 + PSU 115, clearance 3, upper plate 1.6, rail 7.5 + clip 10 + drivers 68, wire room 35, lid ring 20 = 279.4. The base plate lies under the frame, so the bottom ring costs no height. A taller PSU moves the mid ring up and shrinks the wire room (15 left with a 135 mm PSU).

Build and access

Two modules, like a platform-framed house. Lower: base plate, bottom ring and lower posts; stages 3 and 4 run on it alone. Upper: mid ring, upper plate, upper posts and lid ring, built and wired on the bench, then set on 4 studs in the lower posts that locate the decks. Inter-deck wires cross on pluggable terminals. Frame: sheet 4.

Heat and cables

6 mm between drivers for air. The PSU fan is on its top face, 3 mm under the upper plate: widen that gap to a quarter of the fan diameter once it is measured, and vent the lower right panel. Cables leave through four 50 mm grommets: upper left, front and right, and lower left; mains at the back wall beside X0. Leaving the box: Mega USB to the Pi, and J11 plus a fused 24 V feed to the ODrive in the spincoater base.

Sizes: along x across x high (mm)

TB6600 28 x 96 x 68, measured. From the listings: K1 HC1-63 36 x 82 x 65; K2 BMC7-25U 36 x 85 x 68; FB1 / FB2 93 / 153 x 48 x 43; uxcell bar 20 x 78 x 53; SSR 6.1 x 88 x 71 (41F-1Z socket); UK 2.5 B 6.2 x 42.5 x 42; 5 V PSU 76 x 76 x 32; 6 V 63 x 58 x 23; Mega 99 x 53 (A000067). Assumed (dashed): 24 V PSU body 215 x 115 x 50, F0, end clamps, PE blocks, shield clamp, Mega height. Heights add the rail.

At true size (2026-10-17)

The six SSR sockets are 88 deep: they sit on the drivers' rail, not the 65 mm logic row. Terminals are 6.2 wide, not 5.2: the 0V-S groups moved beside the Mega (sheet 2, rail 4), so the field rail ends clear of the upper right grommet. Slot 2 opens into the 110 VAC compartment: DC leads only.

Packing study

Sheet 3. Legal, landscape, 100 %. Scale 1:3.5.
Redrawn 2026-10-17 with every part at its real size; 1/16 in decks, acrylic walls.
Frame and plates: sheet 4. Measure dashed sizes.